Metadata for:

High-throughput synthesis mapping of synthesis and transformation reactions of cesium lead bromide nanocrystals

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Description:

These experiments are a dataset of absorption and photoluminescence emission spectra of cesium lead bromide nanocrystals synthesized under different conditions, published as a companion dataset for the article "Elucidating the weakly reversible Cs-Pb-Br perovskite nanocrystal reaction network with high-throughput maps and transformations" (submitted, DOI link forthcoming). It also includes deconvolution of the absorption spectra and other calculations.

In the mapping experiments, dilute oleylamine, oleic acid, lead and cesium precursor solutions are mixed in a vial and the reaction is initiated by adding benzoyl bromide in a procedure derived from the synthesis reported by Imran et al.[1] In the transformation experiments, reactions between different products are probed by adding more reagents to some of the samples. The synthesis is carried out on a Hamilton NIMBUS4 Microlab liquid handling robot, and the synthesis solutions are transferred to a quartz microplate, diluted by a factor of 10, then absorbance and emission spectra are measured on a Biotek Dyna 2 optical microplate reader, with a volume of 300 uL and a path length of 27.5 mm. Reactions with failed benzoyl bromide injection were excluded from the dataset. All data presented in this work is provided in two datafiles: The mapping experiments are saved in CsPbBr_Synth.csv and transformation experiments are saved in CsPbBr_Trans.csv. Each row is data collected about an individual synthesis, marked with the plate name and row and column identifier used in the automated synthesis and spectral analysis (A1-H12). Differences between different batches (plates) are marked in Table 1. Column entries are explained in Table 2.

Abbreviations:

OLA = Oleylamine
OA = Oleic Acid
Pb = Lead Oleate solution
Cs = Cesium Oleate solution
Br = Benzoyl Bromide solution
T = temperature
rpm = rotations per minute
[OLA],[OA],[Pb],[Cs],[Br] = concentration of solution, in terms of reagent.

$[PbBr_x]$ = Lead bromide complex

1 ML = OLA_2PbBr_4, a single layer lead bromide nanosheet

2 ML = $OLA_2CsPb_2Br_7$, a bilayer lead bromide nanoplate with one layer of intercalated Cs

3 ML = $OLA_2Cs_2Pb_3Br_{10}$, a trilayer lead bromide nanoplate with two layers of intercalated Cs

4 ML = $OLA_2Cs_3Pb_4Br_{13}$, a quadruple layer lead bromide nanoplate with three layers of intercalated Cs

Index	Date	Experiment Type	Overview of experiments performed Solutions added (in µL)	Other parameters	Successful Reactions, Notes
p0	05/22/2019	Map	OLA,OA= 60, Pb = [40,50,60,80],Cs = [20,25,30,51] Br = [40,50,60,80]	T = 30 °C, stir rate = 800 rpm,	83/96
p1	05/22/2019	Map	OA,Br = 60, Pb = [60,80,100,120], Cs = [20,25,30,51], OLA = [30,40,50,60]	T = 30 °C, stir rate = 600 rpm	79/96
p2	06/13/2019	Map/ Transition	OA,Br = 60, expanded p1 to Cs = [40,60], Pb = 140, Pb/Cs transitions	T = 30 °C, stir rate = 600 rpm	96/96
p3	06/13/2019	Map/ Transition	Br=60,Cs =30, OA = [20,40,60,80], Pb = [60,80,100,120], OLA = [30,40,50,60], Pb/Cs transitions	T = 30 °C, stir rate = 600 rpm	95/96
p4	06/17/2019	Map	OA,Br = 60, Pb = [60,80,100,120], Cs = [20,25,30,51], OLA = [30,40,50,60]	T = 40 °C, stir rate = 600 rpm	96/96
p5	07/17/2019	Map	OA,Br = 60, Pb = [60,80,100,120], Cs = [20,25,30,51], OLA = [30,40,50,60]	T = 35 °C, stir rate = 400 rpm	90/96
p6	07/17/2019	Map/ Transition	Expanded p1/p2 to OLA = 20, Cs/Pb/OLA transitions	T = 30 °C, stir rate = 400 rpm	96/96
p7	07/25/2019	Map	OA,Br = 60, Pb = [60,80,100,120], Cs = [20,25,30,51], OLA = [30,40,50,60]	T = 50 °C, stir rate = 400 rpm	96/96
p8	07/25/2019	Map	OA = 60, Cs = 30, Br = [20,40,60,80], Pb = [60,80,100,120], OLA = [30,40,50,60]	T = 30 °C, stir rate = 400 rpm,	96/96, Br solutions 10x diluted
p9	08/29/2019	Map	Expanded main map to OLA = 80, Pb = 40	T = 30 °C, stir rate = 400 rpm	96/96
p10	09/26/2019	Map	Expanded main map to Pb = 20; Br = 60, Pb = 60, OLA = 30, Cs = [20,25,30,40,51,60], OA = [20,40,80,100]	T = 30 °C, stir rate = 400 rpm	80/96
p11	12/05/2019	Transformation	Mostly 2 ML, some [$PbBr_x$]	T = 30 °C, stir rate = 400 rpm	96/96
p12	12/12/2019	Transformation	4 ML, Cs_4PbBr_6, 1 ML	T = 30 °C, stir rate = 400 rpm	96/96
p13	12/12/2019	Transformation	Transformation to [$PbBr_x$] / plate etching	T = 30 °C, stir rate = 400 rpm	96/96
p14	01/06/2020	Map	Expanded main map to Cs = [0,10]	T = 30 °C, stir rate = 400 rpm	60/96
p15	01/16/2020	Map	Repeat of 01/06/2020	T = 30 °C, stir rate = 400 rpm	96/96
p16	01/29/2020	Transformation	$PbBr_2$, $CsPbBr_3$	T = 30 °C, stir rate = 400 rpm	93/96
p17	01/29/2020	Transformation	3 ML, Cs_4PbBr_6	T = 30 °C, stir rate = 400 rpm	95/96
p18	02/06/2020	Transformation	3 ML, $CsPb_2Br_5$	T = 30 °C, stir rate = 400 rpm	96/96

p19	02/06/2020	Transformation	1 ML, $PbBr_2$	T = 30 °C, stir rate = 400 rpm	96/96
p20	02/13/2020	Transformation	3 ML, $CsPb_2Br_5$	T = 30 °C, stir rate = 400 rpm	73/96
p21	02/20/2020repeated 02/27/2020	Map	Expand main map to Cs = [70,80,] Pb = [160,180]	T = 30 °C, stir rate = 400 rpm	96/96
p22	02/20/2020repeated 02/25/2020	Map	Expand main map to Cs = [70,80,] Pb = [160,180]	T = 30 °C, stir rate = 400 rpm	96/96
p23	02/26/2020	Transformation	$[PbBr_x]$, Cs_4PbBr_6	T = 30 °C, stir rate = 400 rpm	96/96
p24	02/26/2020	Transformation	1 ML, 3 ML, $CsPb_2Br_5$	T = 30 °C, stir rate = 400 rpm	96/96
p25	03/05/2020	Transformation	3 ML, $CsPb_2Br_5$	T = 30 °C, stir rate = 400 rpm	96/96

Table 1: Metadata on batches of different experiments. Indices are the leading two or three characters of indices of rows in the dataset.

Column entry	Synth	Trans	Explanation	Data Origin
‘250’-’700’ by 1	Yes	Yes	Measured optical density associated with the specified wavelength	Measured (Biotek Dyna 4)
‘e400.0’ - ‘e550.0’ by 1.0	Yes	Yes	Measured emission counts associated with the specified wavelength, excitation at 350 nm	Measured (Biotek Dyna 4)
'Dodecane', 'OLA', 'Pb', 'OA', 'Cs', 'Br'	Yes	Yes	Volume in µL of reagent solution added to each vial	Controlled (NIMBUS)
'Temp'	Yes	Yes	Synthesis Temperature in °C	Controlled (NIMBUS)
'OLA2', 'Pb2', 'Cs2', 'Dodecane2', 'OA2',	No	Yes	Volume in of reagent solution added to each vial after Br injection	Controlled (NIMBUS)
'tOLA', 'tPb', 'tOA', 'tCs', 'tDodecane',	No	Yes	Total volume in µL of each reagent solution added to vial	Calculated (Direct)
'rcts'	No	Yes	Total volume in µL of reagent solutions added after bromide precursor injection	Calculated (Direct)
'[OLA]', '[Pb]', '[OA]', '[Cs]', '[Br]'	Yes	Yes	Concentration in mM of reagent in vial	Calculated (Direct)
'[OLA2]', '[tOLA]', '[Pb2]', '[tPb]', '[OA2]', '[tOA]', '[Cs2]', '[tCs]'	No	Yes	Concentration in mM of additional reagent and total reagent added to vial after bromide precursor injection	Calculated (Direct)
'StartingMateral', 'Transformation', 'DirectSynthesis'	No	Yes	String marking which function this vial is serving for which reaction. Reactions are numbered 0-148, multiple marks per vial are possible	Calculated (Indirect)
'Deconvolution R^2', 'alpha', 'Num_iters'	Yes	Yes	R^2 of final deconvolution, final penalty value alpha and number of iterations used during fitting.	Parameters extracted during fitting process
'PbBr3-', 'Cs4PbBr6', 'CsPb2Br5', 'PbBr2', '1 ML', '1 ML_2', '2 ML', '2 ML_2', '3 ML', '4 ML', 'CsPbBr3', 'bg', 's1', 's2', 's3', 's4', 's5',	Yes	Yes	Absolute amount of each reference spectrum or background deconvoluted from the measured absorption spectrum. This correlates with optical density of the species, and thus with concentration.	Fitted to reference patterns
'fPbBr3-', 'fCs4PbBr6', 'fCsPb2Br5', 'fPbBr2', 'f1 ML', 'f1 ML_2',	Yes	Yes	Relative amount of each reference spectrum deconvoluted from the measured absorption spectrum. This correlates with the fraction	Calculated from fit to reference patterns

'f2 ML', 'f2 ML_2', 'f3 ML', 'f4 ML', 'fCsPbBr3', 'fbg',			of absorption of a species, and thus with spectral purity or relative yield.	
't1 ML', 't2 ML', 'ft1 ML', 'ft2 ML'	Yes	Yes	Cumulative amount from both reference spectra used for 1 ML and 2 ML plates, respectively	Calculated from fit to reference patterns

Table 2: Metadata on the columns of the dataset.

To help others explore this dataset, we have provided a jupyter notebook file (Data Processing Notebook example.ipynb) that includes the functions used in this analysis and the data selection and visualization procedures used to generate Figure 3 C & D, Figure S 3.7, and Figure 4 in the corresponding scientific publication under xx. The following packages and versions were used to run the notebook: Anaconda 2019.03, conda 4.8.3,[2] numpy 1.18.1,[3] pandas 1.0.1,[4] matplotlib 3.1.3,[5] scipy 1.4.1.[6] The deconvolution function was run with joblib 0.14.1 and scikit-learn 0.22.1.[7]

References:

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